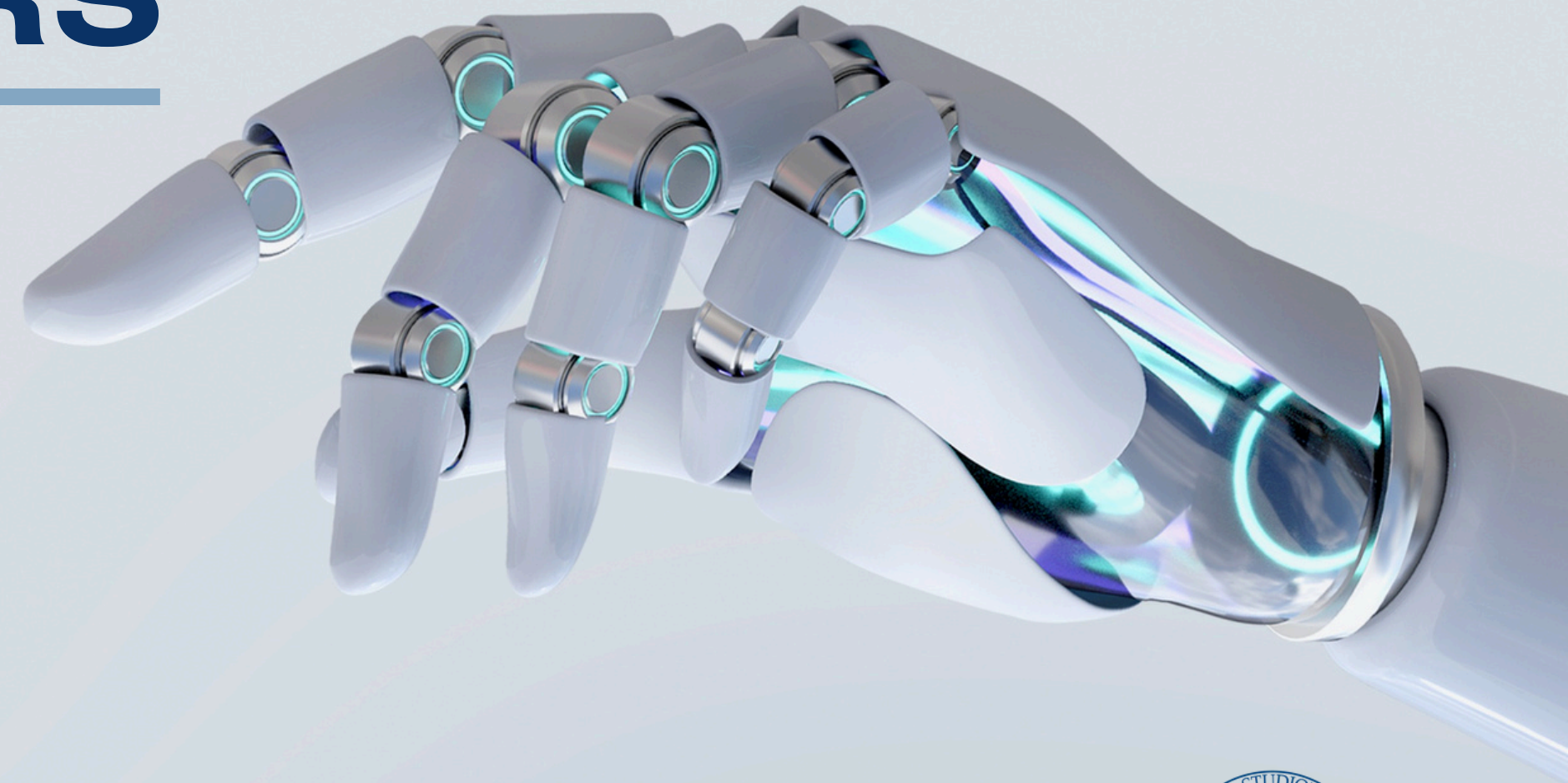


DETECTION OF NON-AUTHENTIC SCIENTIFIC PAPERS

SClgen Case and More Advanced
Generating Methods



ANALYTICS AND DATA SCIENCE FOR ECONOMICS AND MANAGEMENT
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DETECTING SCIGEN GENERATED PAPERS

DATASET CONSTRUCTION

FAKE PAPERS

Source: SCIdgen

Data: 100

SCIdgen is a program that generates random Computer Science research papers. It uses a hand-written context-free grammar.

The articles available are all by a fictitious author: **Ike Antkare**.

The site is no longer working.

Generate a Random Paper

Want to generate a random CS paper of your own? Type in some optional author names below, and click "Generate".

Author 1:
Author 2:
Author 3:
Author 4:
Author 5:

Generate

Reimposta

SCIdgen currently supports Latin-1 characters, but not the full Unicode character set.

DATASET CONSTRUCTION

TRUE PAPERS

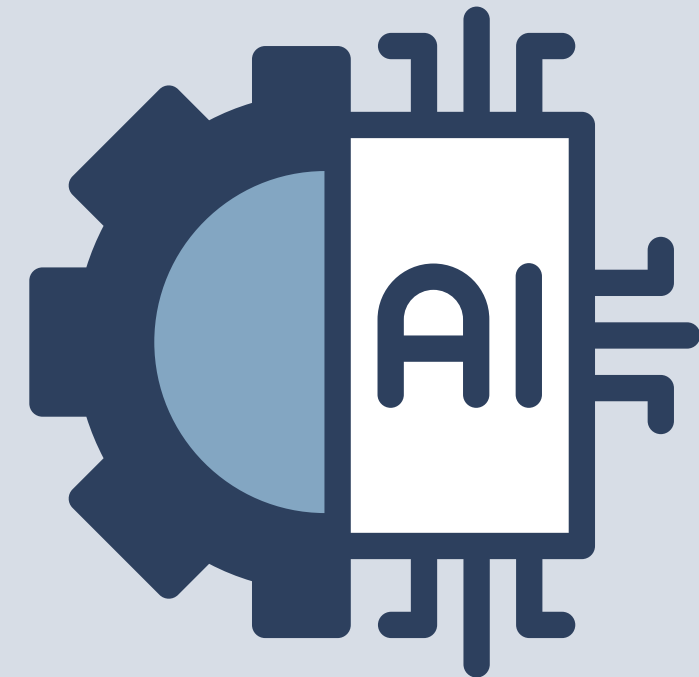
Source: arXiv

Data: 100

arXiv is a free distribution service and an open-access archive for nearly 2.4 million scholarly articles in different fields.

The chosen field is **Computer Science:**

- Networking and Internet Architecture
- Distributed, Parallel, and Cluster Computing
- Operating Systems



TEXT PRE-PROCESSING

NORMALIZATION

Lowercasing
Removal of digits
Punctuation
Special characters

TOKENIZATION

Word segmentation
with spaCy

STOPWORDS

Removed with NLTK
+ spaCy
Custom stopwords:
"Ike", "Antkare", "Ike
Antkare"

LEMMATIZATION

Words reduced to
their base form
(e.g., running → run)

TF - IDF

TERM FREQUENCY – INVERSE DOCUMENT FREQUENCY

After pre-processing, each document was transformed into a **vector representation** using **TF - IDF** → suitable format for machine learning models.

TF - IDF balances:

- The **local importance** of a term within a document (TF)
- The **global rarity** across all documents (IDF)

CLASSIFICATION MODELS

1

SUPPORT VECTOR MACHINE

Robust in high-dimensional spaces and widely used in text classification.

2

LOGISTIC REGRESSION

Simple and interpretable linear model and effective with sparse TF-IDF vectors.

3

NAÏVE BAYES

Classic probabilistic model. Well-suited for text data despite its independence assumption.

4

RANDOM FOREST

Ensemble method that captures nonlinear patterns and reduces overfitting through decision tree averaging.

RESULTS

SUPPORT VECTOR MACHINE - L2

Training set (60%) → 120 obs

Test set (40%) → 80 obs

CONFUSION MATRIX AND CLASSIFICATION REPORT

Perfect
separation

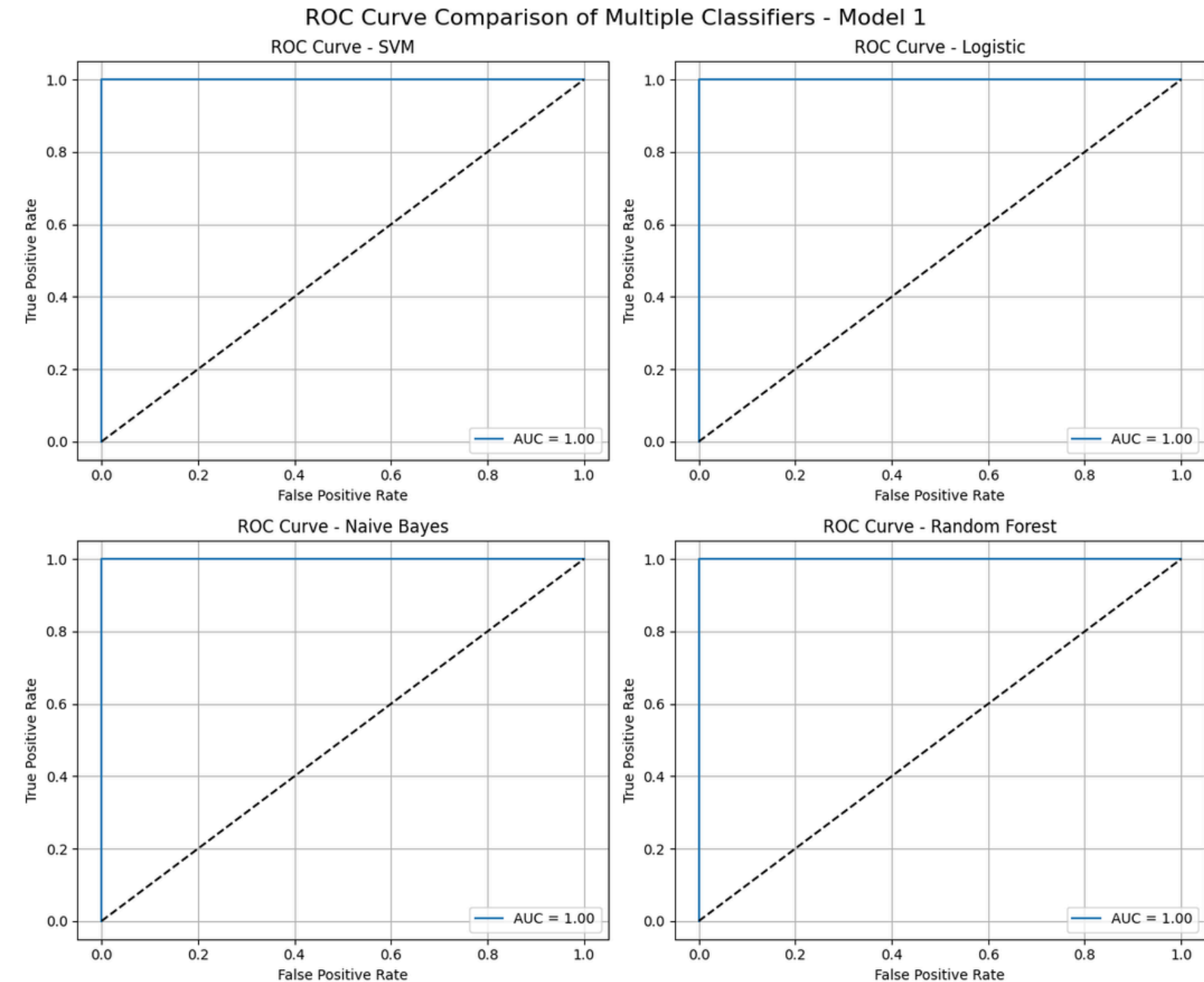
	Predicted 0	Predicted 1
True 0	40	0
True 1	0	40

Class	Precision	Recall	F1-score	Support
0	1.00	1.00	1.00	40
1	1.00	1.00	1.00	40
Accuracy			1.00	80
Macro avg	1.00	1.00	1.00	80
Weighted avg	1.00	1.00	1.00	80

ROC CURVES - AUC

The figure shows the ROC curves of the four classifiers.

All models achieve an **AUC of 1.0**, which indicates a perfect ability to distinguish between authentic and fake papers.



N-GRAMS: PATTERN RECOGNITION

2-5 N-GRAM ANALYSIS

Rank	N-gram (Real)	Weight	N-gram (Fake)	Weight
1	vol pp	0.208126	<u>proceeding</u> workshop	-0.268824
2	ipv address	0.123427	<u>pro</u> ceeding	-0.246797
3	batch size	0.122926	<u>proceeding</u> conference	-0.228366
4	high performance	0.115872	consider harmful	-0.219000
5	ue antenna	0.104470	gigabit switch	-0.191527
6	quantum network	0.097916	context free	-0.187862
7	international conference	0.095163	cache coherence	-0.182612
8	real world	0.091634	technical report	-0.176043
9	end end	0.089868	context <u>free</u> grammar	-0.156652
10	control system	0.089447	<u>free</u> grammar	-0.156652
11	network traffic	0.089169	wide area	-0.156114
12	time slot	0.086644	<u>proceeding</u> fpca	-0.151975
13	ip address	0.079510	digital analog	-0.150075
14	large batch	0.079330	boolean logic	-0.148321
15	cross chain	0.077387	area network	-0.130890

Fake papers often display redundant structures, easier to detect.

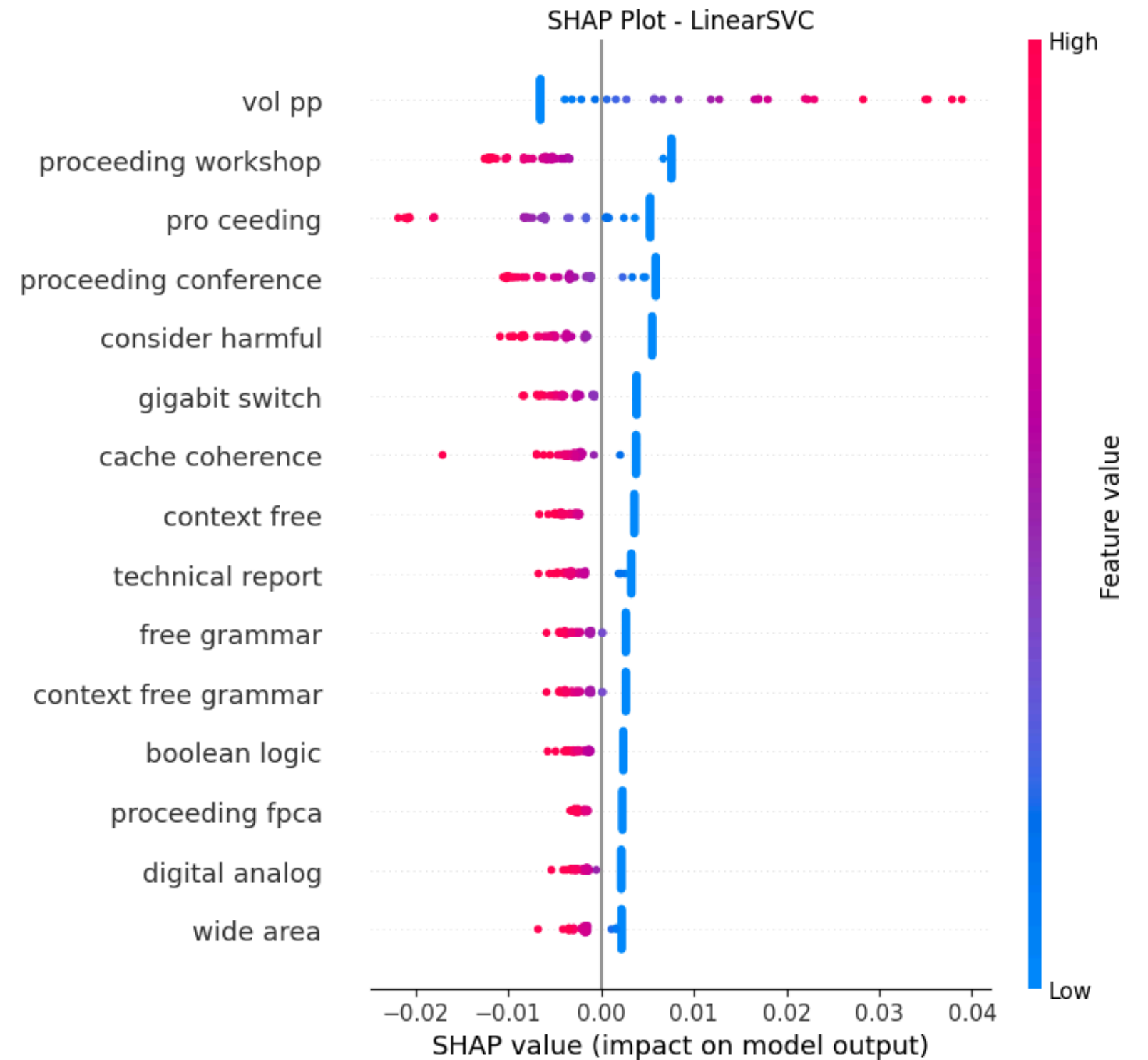
- **Weight > 0:** pushes towards class 1
- **Weight < 0:** pushes towards class 0

→ vol pp stands for volume and pages

SHAP METHOD

SHAPLEY ADDITIVE EXPLANATIONS

SHAP analysis confirms that the most important features for the model are the same identified in the n-gram analysis.



DETECTING ADVANCED GENERATED PAPERS

DATASET CONSTRUCTION

MODEL 1

SCIGEN + CHATGPT

SClgen articles modified with ChatGPT to improve structure and patterns.

10

LLM

Academic-style documents generated with the open-source TinyLlama model.

35

ARXIV

Authentic scientific articles downloaded from arXiv.

45

RESULTS

SUPPORT VECTOR MACHINE - L2

Training set (60%) → 54 obs

Test set (40%) → 30 obs

CONFUSION MATRIX AND CLASSIFICATION REPORT

Accuracy drops from 100% to 89%

This decline reflects the closer similarity of LLM text to authentic papers, making detection harder.

	Pred. 0	Pred. 1
True 0	16	2
True 1	2	16

Class	Precision	Recall	F1-score	Support
0	0.89	0.89	0.89	18
1	0.89	0.89	0.89	18
Accuracy			0.89	36
Macro avg	0.89	0.89	0.89	36
Weighted avg	0.89	0.89	0.89	36

OTHER CLASSIFICATION METHODS

LOGISTIC REGRESSION

	Pred. 0	Pred. 1
True 0	14	4
True 1	2	16

Class	Precision	Recall	F1-score	Support
0	0.88	0.78	0.82	18
1	0.80	0.89	0.84	18
Accuracy			0.83	36
Macro avg	0.84	0.83	0.83	36
Weighted avg	0.84	0.83	0.83	36

Accuracy drops from
100% to 83%

NAÏVE BAYES

	Pred. 0	Pred. 1
True 0	17	1
True 1	2	16

Class	Precision	Recall	F1-score	Support
0	0.89	0.94	0.92	18
1	0.94	0.89	0.92	18
Accuracy			0.92	36
Macro avg	0.92	0.92	0.92	36
Weighted avg	0.92	0.92	0.92	36

Accuracy drops from
100% to 92%

OTHER CLASSIFICATION METHODS

RANDOM FOREST

	Pred. 0	Pred. 1
True 0	18	0
True 1	0	18

Class	Precision	Recall	F1-score	Support
0	1.00	1.00	1.00	18
1	1.00	1.00	1.00	18
Accuracy			1.00	36
Macro avg	1.00	1.00	1.00	36
Weighted avg	1.00	1.00	1.00	36

Accuracy remains at 100%

Although the results appear more promising, the **limited dataset size** represents a significant limitation.

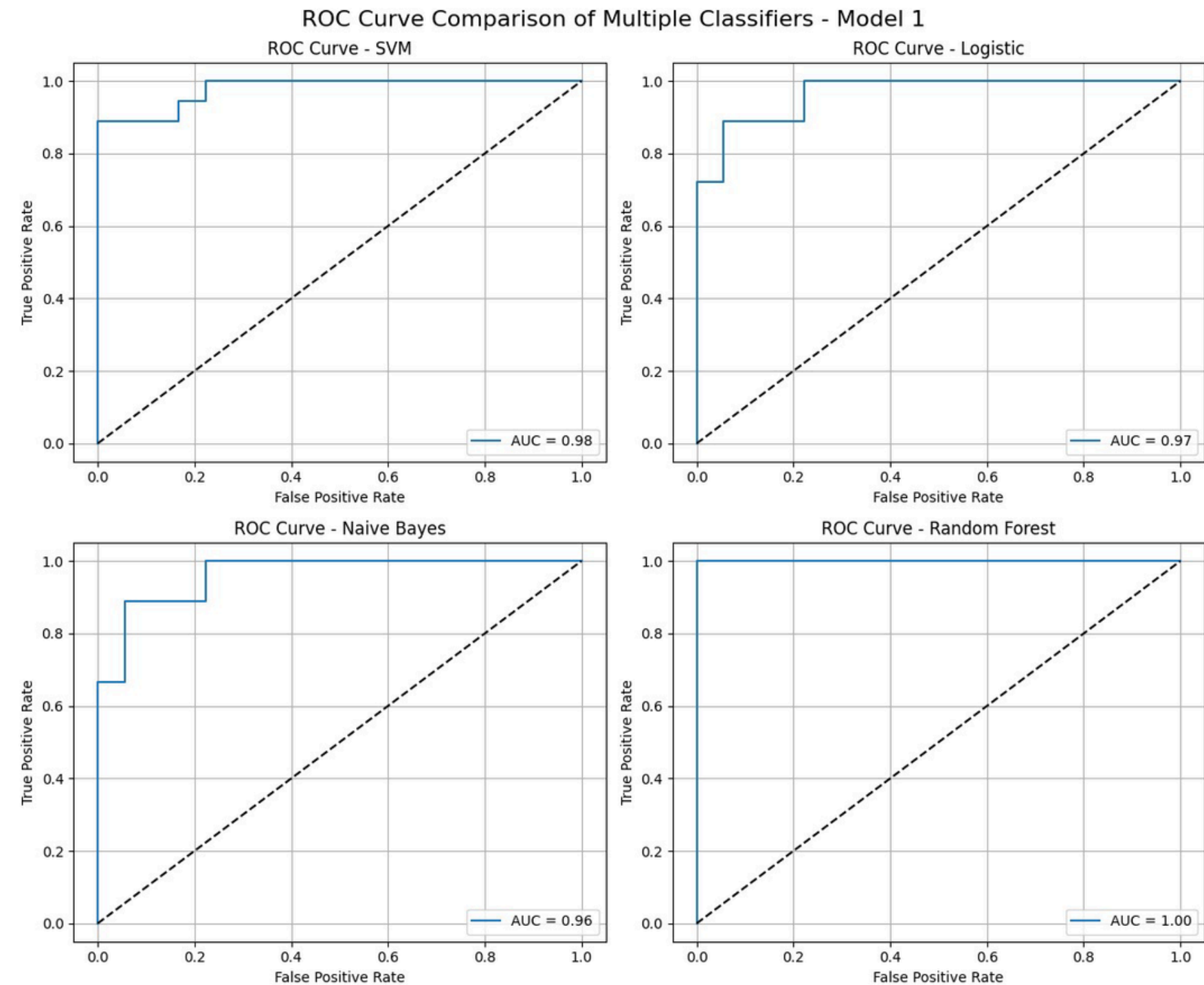
36 test observations

ROC CURVES - AUC

The figure shows the ROC curves of the four classifiers.

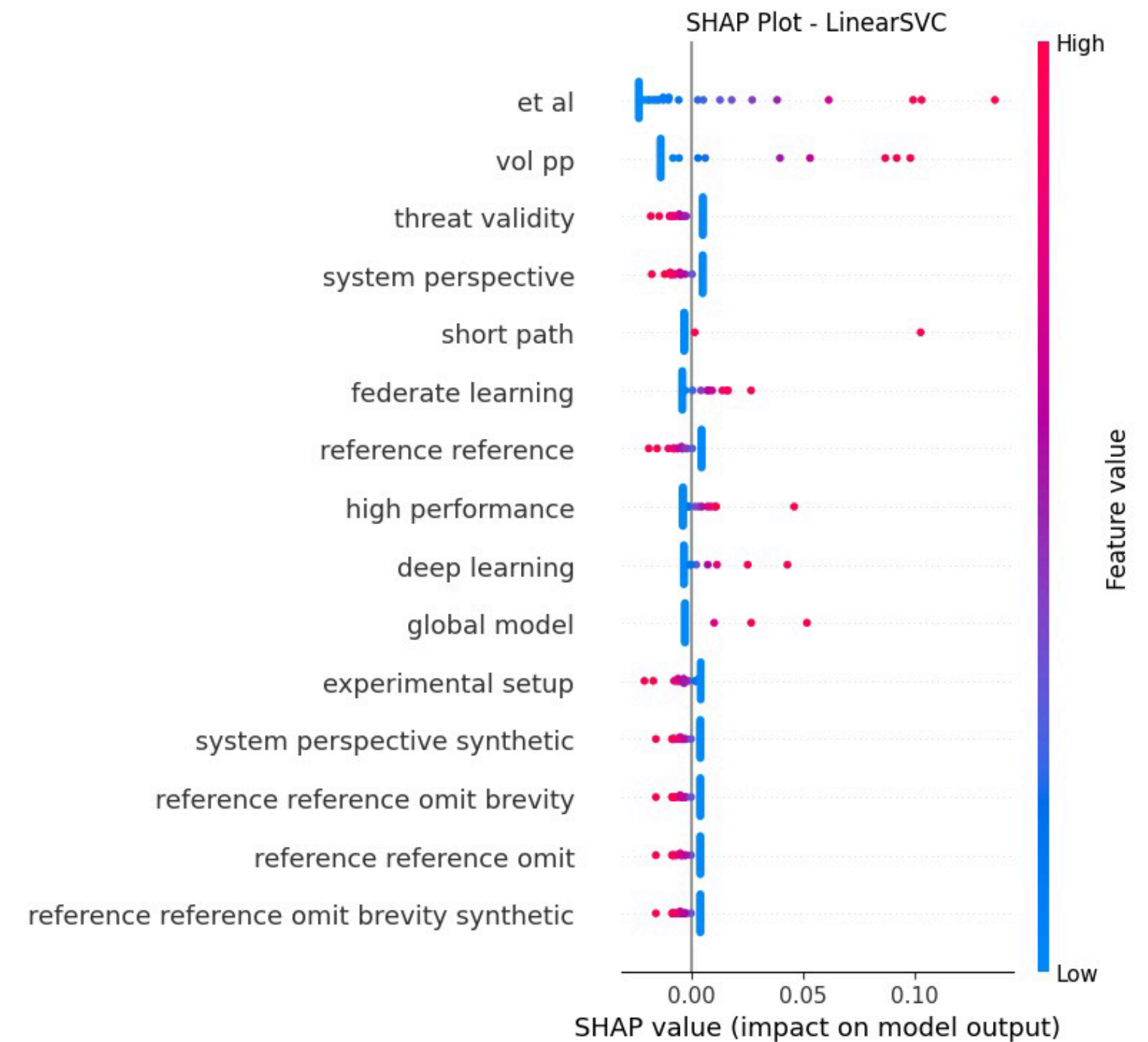
Slightly lower AUC value reflects an increased challenge in classification, even if the overall performance of the models remains strong.

→ AUC 0.96 - 1.00



N-GRAMS AND SHAP ANALYSIS

Rank	N-gram (Real)	Weight	N-gram (Fake)	Weight
1	et al	0.467073	mixed precision	-0.366800
2	vol pp	0.447904	holo artisan	-0.342372
3	batch size	0.353157	threat validity	-0.284088
4	online available	0.316916	system perspective	-0.278062
5	bad case	0.286687	reference <u>reference</u>	-0.242812
			omit brevity	
6	smart contract	0.286330	distribute system	-0.242812
			september abstract	
7	step size	0.284732	synthetic author	-0.242812
			institute <u>distribute</u>	
			system	
8	formal specification	0.245368	synthetic document	-0.242812
9	large batch	0.242016	institute <u>distribute</u>	-0.242812
			system	
10	top top	0.230980	institute <u>distribute</u>	-0.242812
			system september	
11	international conference	0.221889	system september	-0.242812
12	global model	0.216582	institute <u>distribute</u>	-0.242812
			system september	
			abstract	
13	arxiv preprint	0.206887	author institute	-0.242812
			distribute system	
			september	
14	preprint arxiv	0.206887	<u>reference</u> <u>reference</u>	-0.242812
15	arxiv preprint arxiv	0.206887	<u>reference</u> <u>reference</u>	-0.242812
			omit	



DATASET CONSTRUCTION

MODEL 2

SCIGEN + CHATGPT

SClgen articles modified with ChatGPT to improve structure and patterns.

10

LLM

Academic-style documents generated with the open-source TinyLlama model.

35

LLM + ARXIV

Authentic arXiv papers progressively noised until they look fakes.

45

ARXIV

Authentic scientific articles downloaded from arXiv.

90

RESULTS

SUPPORT VECTOR MACHINE - L2

Training set (60%) → 108 obs

Test set (40%) → 72 obs

CONFUSION MATRIX AND CLASSIFICATION REPORT

Accuracy drops from 100% to 93%

The model is **less trivial**,
fake papers also include corrupted
real documents.

	Pred. 0	Pred. 1
True 0	33	3
True 1	2	34

Class	Precision	Recall	F1-score	Support
0	0.94	0.92	0.93	36
1	0.92	0.94	0.93	36
Accuracy			0.93	72
Macro avg	0.93	0.93	0.93	72
Weighted avg	0.93	0.93	0.93	72

OTHER CLASSIFICATION METHODS

LOGISTIC REGRESSION

	Pred. 0	Pred. 1
True 0	31	5
True 1	1	35

Class	Precision	Recall	F1-score	Support
0	0.97	0.86	0.91	36
1	0.88	0.97	0.92	36
Accuracy			0.92	72
Macro avg	0.92	0.92	0.92	72
Weighted avg	0.92	0.92	0.92	72

Accuracy drops from
100% to 92%

NAÏVE BAYES

	Pred. 0	Pred. 1
True 0	33	3
True 1	5	31

Class	Precision	Recall	F1-score	Support
0	0.87	0.92	0.89	36
1	0.91	0.86	0.89	36
Accuracy			0.89	72
Macro avg	0.89	0.89	0.89	72
Weighted avg	0.89	0.89	0.89	72

Accuracy drops from
100% to 89%

OTHER CLASSIFICATION METHODS

RANDOM FOREST

	Pred. 0	Pred. 1
True 0	36	0
True 1	0	36

Class	Precision	Recall	F1-score	Support
0	1.00	1.00	1.00	36
1	1.00	1.00	1.00	36
Accuracy			1.00	72
Macro avg	1.00	1.00	1.00	72
Weighted avg	1.00	1.00	1.00	72

Accuracy remains at 100%

Note: publications on arXiv are updated daily.

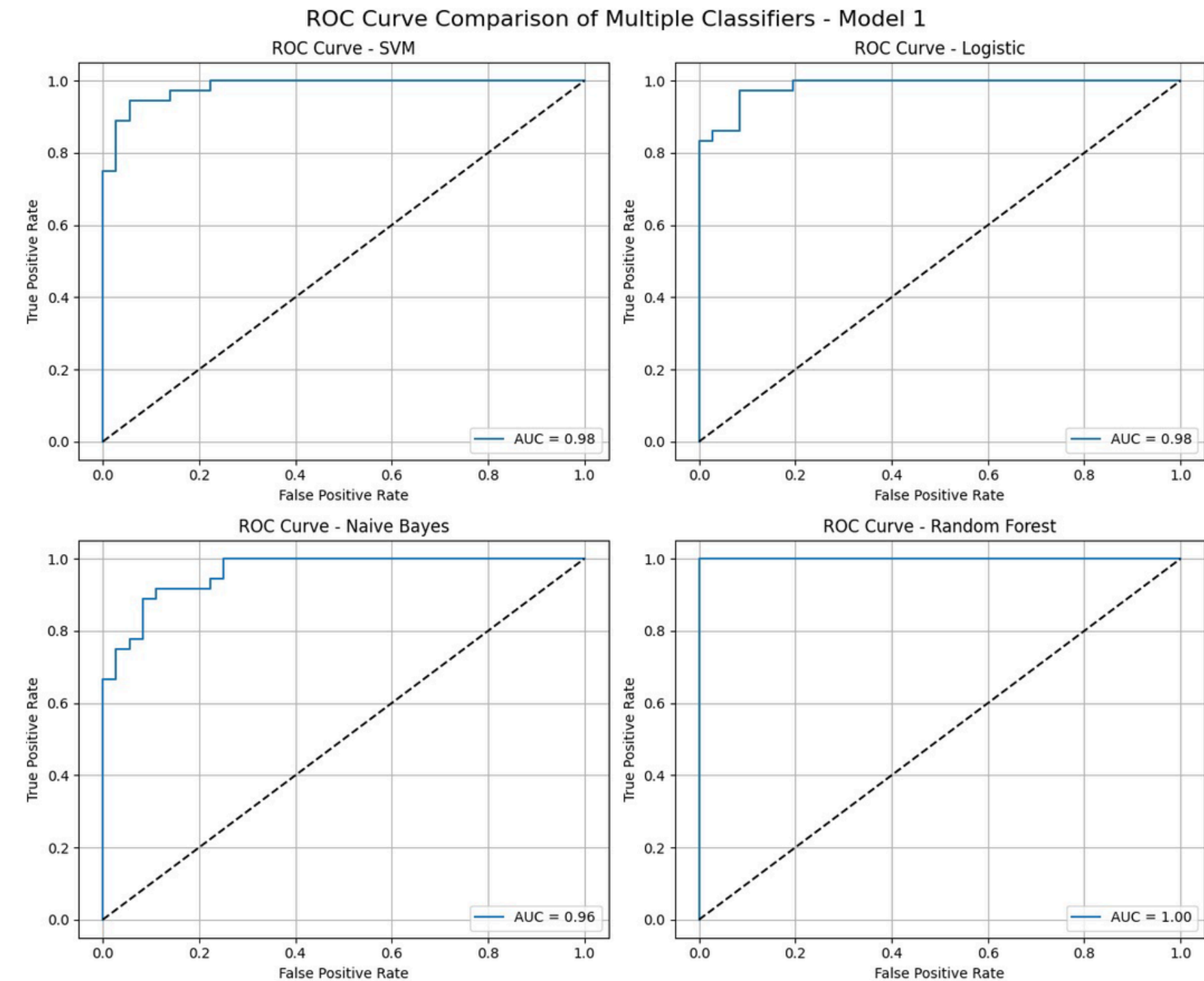
Results of the analysis may differ slightly when the code is sent again.

ROC CURVES - AUC

The figure shows the ROC curves of the four classifiers.

Random Forest still shows strong robustness.

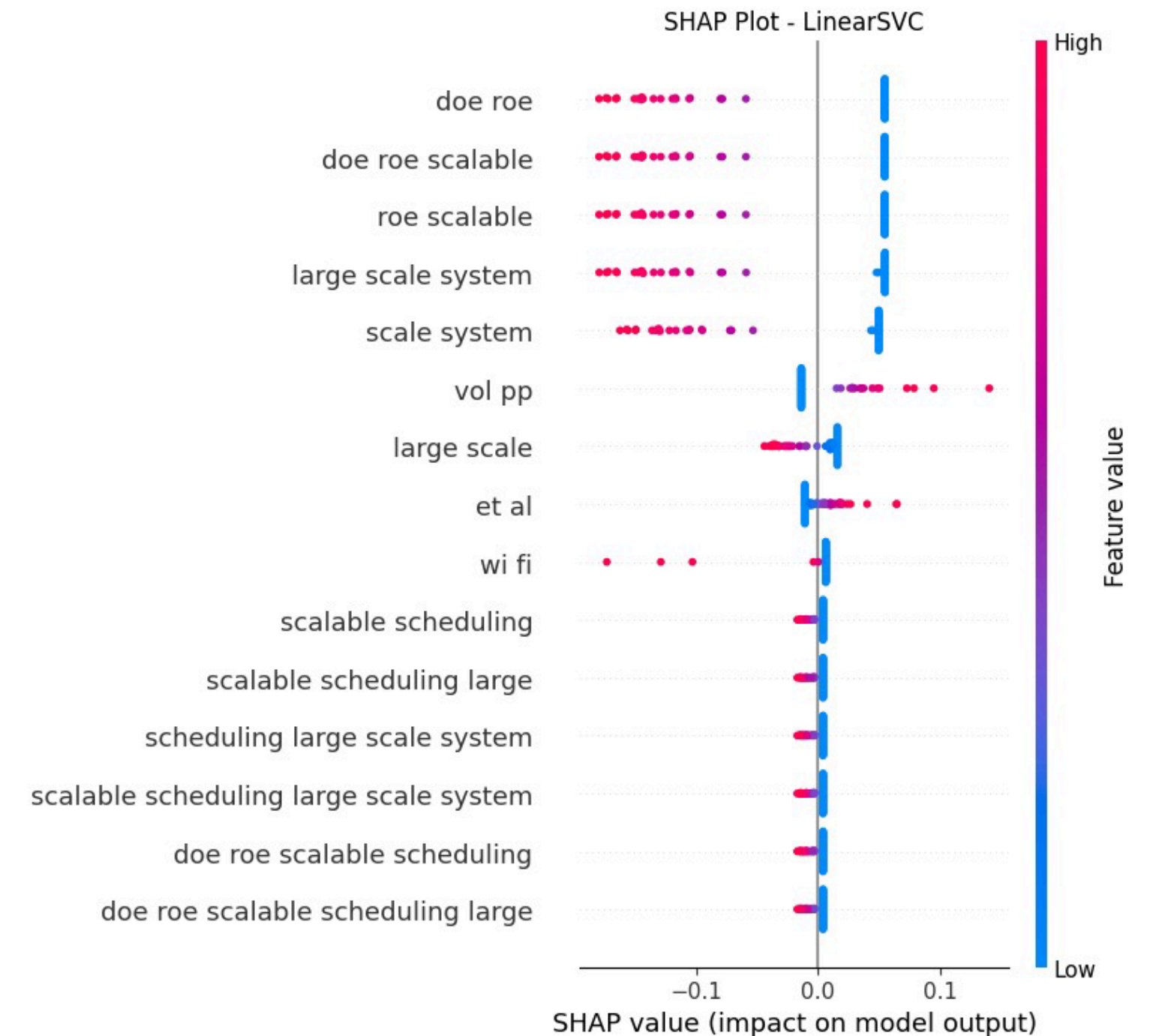
Naïve Bayes performs slightly worse due to its independence assumption.



AUC 0.94 - 1.00

N-GRAMS AND SHAP ANALYSIS

Rank	N-gram (Real)	Weight	N-gram (Fake)	Weight
1	vol pp	0.514356	doe roe scalable	-0.762000
2	et al	0.311016	doe roe	-0.762000
3	international conference	0.274895	roe scalable	-0.762000
4	leaf switch	0.253222	large scale system	-0.762000
5	proceeding th	0.248256	scale system	-0.724525
6	porto alegre	0.246575	threat validity	-0.348859
7	online available	0.229068	holo artisan	-0.346426
8	share memory	0.220238	experimental setup	-0.345480
9	ieee transaction	0.186775	system perspective	-0.344376
10	rio grande	0.184166	reference reference	-0.330951
11	preprint arxiv	0.183555	large scale	-0.314842
12	rio grande sul	0.181577	system perspective	-0.311172
13	grande sul	0.181577	synthetic author	-0.311172
14	two way	0.179981	omit brevity synthetic	-0.311172
15	arxiv preprint	0.176071	system perspective synthetic	-0.311172
			perspective synthetic author	-0.311172



“doe roe” stand for fake authors,
“John Doe” and “Jane Roe”

CONCLUSIONS

1

SClgen alone makes fake papers easy to detect, but introducing noise or **LLM** generated text reduces accuracy for simpler models.

2

Random Forest is robust: tree-based models maintain high performance even with more realistic fake texts.

3

N-gram analysis shows that fake texts contain nonsensical word combinations, helping to interpret model decisions.

FUTURE DEVELOPMENTS

1

Larger datasets including more authentic and LLM generated papers to improve generalization.

2

Advanced feature engineering like POS tags, word embeddings, BERT or other contextual representations.

3

Cross-domain detection extending the study to papers from different scientific fields.

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THANKS FOR YOUR ATTENTION

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